

The Erdős–Szekeres Conjecture Revisited

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Abstract

The famous and still open *Erdős–Szekeres Conjecture* from 1935 states that every set of at least $2^{k-2} + 1$ points in the plane with no three being collinear contains k points in *convex position*, that is, k points that are vertices of a convex polygon. In this paper, we revisit this conjecture and show several new related results.

First, we prove a relaxed version of the Erdős–Szekeres Conjecture by showing that every set of at least $2^{k-2} + 1$ points in the plane with no three being collinear contains a *split k -gon*, a relaxation of k -tuple of points in convex position. Moreover, we show that this is tight, showing that the value $2^{k-2} + 1$ from the Erdős–Szekeres Conjecture is exactly the right threshold for split k -gons.

We obtain an analogous relaxation in a much more general setting of ordered 3-uniform hypergraphs where we also show that, perhaps surprisingly, a corresponding generalization of the Erdős–Szekeres Conjecture is not true. Finally, we prove the Erdős–Szekeres Conjecture for so-called *decomposable sets* and provide new constructions of sets of 2^{k-2} points without k points in convex position, generalizing all previously known constructions of such point sets and allowing us to computationally tackle the Erdős–Szekeres Conjecture for large values of k .

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1 Introduction

A finite set of points on a plane is in *general position* if there are no three points on a single line and in *convex position* if they form the vertices of a convex polygon. We consider finite point sets in general position where no two points have the same x - or y -coordinate.

Motivated by a problem posed by Esther Klein, Erdős and Szekeres [7] showed that for any positive integer k there is a minimum integer $ES(k)$ such that any set of $ES(k)$ points in the plane in general position contains k points in convex position. This famous result, called the *Erdős–Szekeres Theorem*, was a starting point of both discrete geometry and Ramsey theory and was obtained as one of the first applications of Ramsey’s Theorem, which was independently discovered by Erdős and Szekeres [7] in the same paper. Ramsey’s Theorem, however, yields a poor bound on the minimum size $ES(k)$ of a planar point set in general position that guarantees the existence of k points in convex position. Thus, Erdős and Szekeres [7] also provided an alternative proof that yields a much better estimate on $ES(k)$. This alternative argument uses more geometry and is based on so-called *caps* and *cups*.



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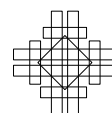
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For a positive integer a , we say that a points in the plane form an a -cap if they lie on a graph of a concave function. Similarly, for $u \in \mathbb{N}$, a set of u points in the plane forms a u -cup if its points lie on a graph of a convex function. Using this notion, Erdős and Szekeres [7] proved the following result known as the *Erdős–Szekeres Cap-Cup Theorem*.

► **Theorem 1** (The Erdős–Szekeres Cap-Cup Theorem [7]). *For all integers $a, u \geq 2$, every set of at least $\binom{a+u-4}{a-2} + 1$ points in the plane in general position contains an a -cap or a u -cup.*

Moreover, this is tight. That is, for all integers $a, u \geq 2$, there are sets of $\binom{a+u-4}{a-2}$ points in the plane in general position with neither an a -cap nor a u -cup.

Since points of an a -cap and a u -cup are in convex position, Theorem 1 gives

$$ES(k) \leq \binom{2k-4}{k-2} + 1 \quad (1)$$

for any integer $k \geq 2$. In fact, observe that every set of k points in convex position is a union of an a -cap and a u -cup that share common endpoints and satisfy $a + u = k + 2$.

The bound (1) gives roughly $ES(k) \leq 4^k$. Erdős–Szekeres believed that this could be significantly improved and posed the following conjecture.

► **Conjecture 2** (The Erdős–Szekeres Conjecture [7]). *For every integer $k \geq 2$, we have*

$$ES(k) = 2^{k-2} + 1.$$

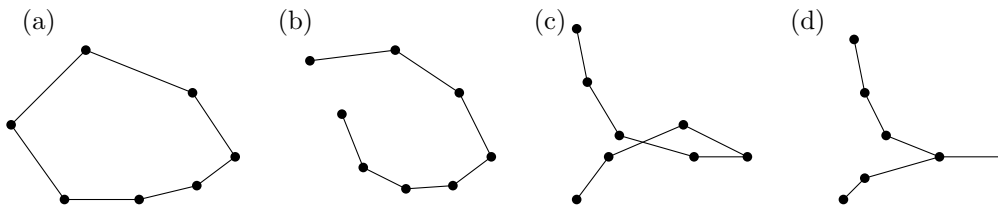
When posed in 1935, the conjecture was known to be true only for $k \leq 4$. Later, it was proved for $k = 5$, and Peters and Szekeres [19] proved it also for $k = 6$ in 2006 using a computer-assisted approach. In the 1960s, Erdős and Szekeres [8] showed $ES(k) \geq 2^{k-2} + 1$ for any integer $k \geq 2$, further supporting the conjecture. Despite many efforts, this conjecture is still open for $k \geq 7$, and Erdős himself offered \$500 reward for its solution. For many decades, the bound (1) was the best known, but then several smaller-order improvements appeared [5, 14, 16, 18, 21, 22]. In 2017, Suk [20] made a breakthrough on this conjecture by proving $ES(k) \in 2^{k+O(k^{2/3} \log k)}$ and his bound was then slightly improved by Holmsen, Mojarrad, Pach, and Tardos [13] to $ES(k) \in 2^{k+O(\sqrt{k} \log k)}$. Although these bounds are fairly close to $2^{k-2} + 1$, there is still a gap between the known lower and upper bounds, and this famous conjecture thus remains open.

2 Our results

In this paper, we introduce a natural relaxed variant of the Erdős–Szekeres Conjecture for which the number $2^{k-2} + 1$ is exactly the right threshold. To our knowledge, this is the first version of this famous conjecture where the conjectured value $2^{k-2} + 1$ appears as the right threshold. We discuss abstract combinatorial extensions of this result and connections to the original Erdős–Szekeres Conjecture, some of which indicate that this conjecture might be false. We also prove the Erdős–Szekeres Conjecture restricted to special point sets, and we provide new examples of 2^{k-2} point sets with no k points in convex position.

2.1 Split polygons

For a positive integer N , let P be a set of N points in the plane in general position. For a positive integer k , a *split k -gon* in P is a set of points from P that is formed by an a -cap and a u -cup that share the rightmost point and that satisfy $a + u = k + 2$; see Figure 1. Note that a split k -gon can contain k or $k + 1$ points and that if the a -cap and the u -cup share also the leftmost point, then the points of the split k -gon are in convex position.



■ **Figure 1** (a–d) Examples of split 7-gons. Note that the split 7-gons in (a) and (d) have only 7 points while the remaining split 5-gons contain 8 points.

We consider a variant of the Erdős–Szekeres Theorem for split polygons. More specifically, for a positive integer k , let $ES_{split}(k)$ be the minimum positive integer N such that every finite set of at least N points in the plane in general position contains a split k -gon. Since $ES_{split}(k) \leq ES(k)$, we have $ES_{split}(k) \leq 2^{(1+o(1))k}$ by the result of Suk [20].

As our first main result, we prove that the numbers $ES_{split}(k)$ attain exactly the value $2^{k-2} + 1$ from the Erdős–Szekeres Conjecture.

► **Theorem 3.** *For every integer $k \geq 2$, we have $ES_{split}(k) = 2^{k-2} + 1$.*

In fact, we can prove a stronger statement in the setting corresponding to a refinement of the Erdős–Szekeres Conjecture as introduced by Erdős, Tuza, and Valtr [9]. In their refinement, one tries to find k -tuples of points in convex position in point sets that do not contain a -caps and u -cups of prescribed sizes a and u .

We state this formally in our setting. For integers $a, u, k \geq 2$, let $ES_{split}(a, u, k)$ be the minimum positive integer N such that every finite set of at least N points in the plane in general position contains an a -cap, a u -cup, or a split k -gon. We are mostly interested in the cases $\max\{a, u\} \leq k \leq a + u - 2$ as otherwise at least one parameter from a, u, k is unnecessary. Therefore, we define

$$T = \{(a, u, k) : a, u \geq 2 \text{ and } \max\{a, u\} \leq k \leq a + u - 2\}.$$

For any triple $(a, u, k) \in T$, we can then determine the numbers $ES_{split}(a, u, k)$ exactly.

► **Theorem 4.** *For every $(a, u, k) \in T$, we have*

$$ES_{split}(a, u, k) = 1 + \sum_{i=k-a+2}^u \binom{k-2}{i-2}.$$

Note that $ES_{split}(k) = ES_{split}(k, k, k)$. Thus, Theorem 3 follows from Theorem 4 by setting $a = u = k$ as then

$$ES_{split}(k) = 1 + \sum_{i=2}^k \binom{k-2}{i-2} = 2^{k-2} + 1.$$

Also, observe that Theorem 1 is a special case of Theorem 4 for $k = a + u - 2$.

We also remark that if $ES(a, u, k)$ is the minimum positive integer N such that every finite set of at least N points in the plane in general position contains an a -cap, a u -cup, or k points in convex position, then it is conjectured that

$$ES(a, u, k) = 1 + \sum_{i=k-a+2}^u \binom{k-2}{i-2} \quad (2)$$

for all $(a, u, k) \in T$. However, this equality is known to be true only in the trivial case $k \geq a + u - 3$ and Baek [1] proved it when a or u equals 4. Surprisingly, Erdős, Tuza, and Valtr [9] showed that the Erdős–Szekeres conjecture is equivalent to (2).

2.2 Abstract split polygons

The proof of the upper bound in Theorem 4 is based on techniques used by Moshkovitz and Shapira [17] when estimating ordered Ramsey numbers of so-called monotone paths. It turns out that the proof can be generalized to this abstract combinatorial setting. To state this extension, we need to introduce some definitions.

For a positive integer N , let $\mathcal{K}_N^3$ be the ordered complete 3-uniform hypergraph with the vertex set $[N] = \{1, \dots, N\}$ ordered by the standard ordering $<$ of the positive integers. A 2-coloring of $\mathcal{K}_N^3$ is a function χ that assigns either a red or blue color to each hyperedge of $\mathcal{K}_N^3$. A monotone path $\mathcal{P}_n^3$ is an ordered 3-uniform hypergraph on $n \geq 2$ vertices where the hyperedges are formed by triples of vertices that are consecutive in the vertex ordering of $\mathcal{P}_n^3$.

Let $P = \{p_1, \dots, p_N\}$ be a set of points in the plane in general position with $x(p_1) < \dots < x(p_N)$ where $x(p_i)$ is the x -coordinate of a point p_i . The 2-coloring induced by P is a 2-coloring χ_P of $\mathcal{K}_N^3$ that colors a hyperedge $\{i, j, k\}$ with $i < j < k$ red if the points p_i, p_j, p_k form a 3-cap and blue if p_i, p_j, p_k form a 3-cup. Observe that, for each n , red monotone paths on n vertices in χ_P correspond to n -caps in P and that blue monotone paths on n vertices in χ_P correspond to n -cups in P .

In 2014, Moshkovitz and Shapira [17] proved the following result, whose first part vastly generalizes the first part of the Erdős–Szekeres Cap-Cup Theorem (Theorem 1) as it deals with all 2-colorings of $\mathcal{K}_N^3$ and not only with those induced by a point set of size N .

► **Theorem 5** ([7, 11, 17]). *For all integers $a, u \geq 2$ and $N \geq \binom{a+u-4}{a-2} + 1$, every 2-coloring of $\mathcal{K}_N^3$ contains either a red copy of $\mathcal{P}_a^3$ or a blue copy of $\mathcal{P}_u^3$.*

Moreover, this is tight. That is, there are 2-colorings of $\mathcal{K}_N^3$ with $N = \binom{a+u-4}{a-2}$ that do not contain a red copy of $\mathcal{P}_a^3$ nor a blue copy of $\mathcal{P}_u^3$.

In other words, Theorem 5 states that the ordered Ramsey number [3, 6] of $\mathcal{P}_a^3$ and $\mathcal{P}_u^3$ equals $\binom{a+u-4}{a-2} + 1$. We extend this result similarly as Theorem 4 extends Theorem 1.

We start by generalizing split polygons. Let χ be a 2-coloring of $\mathcal{K}_N^3$ with colors red and blue. For $k \in \mathbb{N}$, an abstract split k -gon in χ is an ordered subhypergraph of $\mathcal{K}_N^3$ that is formed by a red monotone path $\mathcal{P}_a^3$ and a blue monotone path $\mathcal{P}_u^3$ that share the rightmost vertex and that satisfy $a + u = k + 2$. Observe that an abstract split k -gon can have at most $k + 1$ vertices. Also, note that, for every abstract split k -gon C in a coloring χ_P induced by a point set P , the two monochromatic monotone paths in C can share only their endpoint pair and that abstract split k -gons in χ_P are in one-to-one correspondence with split k -gons in P .

Generalizing the definition of $ES_{split}(a, u, k)$, we let $C_{split}(a, u, k)$ be the minimum positive integer N such that every 2-coloring of $\mathcal{K}_N^3$ with colors red and blue contains a red copy of $\mathcal{P}_a^3$, a blue copy of $\mathcal{P}_u^3$, or an abstract split k -gon. Observe that considering the 2-colorings induced by P gives, for every $(a, u, k) \in T$,

$$ES_{split}(a, u, k) \leq C_{split}(a, u, k). \quad (3)$$

As our next result, we fully determine the numbers $C_{split}(a, u, k)$ by showing that $ES_{split}(a, u, k) = C_{split}(a, u, k)$ for every $(a, u, k) \in T$.

► **Theorem 6.** *For every $(a, u, k) \in T$, we have*

$$C_{split}(a, u, k) = 1 + \sum_{i=k-a+2}^u \binom{k-2}{i-2}.$$

Similarly as before, note that Theorem 5 is a special case of Theorem 6 for $k = a + u - 2$.

2.3 Finding polygons that are not split

The only difference between the setting in the Erdős–Szekeres Conjecture and the setting in Theorem 3 is that in the latter one, we allow the leftmost endpoints of the a -cap and the u -cup to be different. Enforcing these two endpoints to meet without increasing the size of the considered point sets would resolve the Erdős–Szekeres Conjecture. However, the combinatorial structure used to prove Theorem 3 does not seem strong enough to ensure this. Moreover, we will show that this strengthening is not true in the abstract combinatorial setting from Subsection 2.2.

First, we generalize the concept of convex position. For a positive integer N , let χ be a 2-coloring of $\mathcal{K}_N^3$. For a positive integer k , a *weak k -gon* in χ is an ordered subhypergraph of $\mathcal{K}_N^3$ that is formed by a red monotone path $\mathcal{P}_a^3$ and a blue monotone path $\mathcal{P}_u^3$ that share both their end-vertices and satisfy $a + u = k + 2$. That is, a weak k -gon is a particular case of an abstract split k -gon. Note that the red and the blue monotone path can share other vertices, not only their two end-vertices. If we exclude this possibility and allow the two paths to share only their end-vertices, then we call such ordered subhypergraph a *strong k -gon* in χ . If χ_P is a 2-coloring induced by a point set P , then every weak k -gon is a strong k -gon in χ_P and corresponds to a set of k points from P that are in convex position.

Let $C_{\text{weak}}(k)$ be the minimum $N \in \mathbb{N}$ such that every 2-coloring of $\mathcal{K}_N^3$ contains a weak k -gon. An analogous number for strong k -gons is denoted by $C_{\text{strong}}(k)$. Clearly,

$$C_{\text{strong}}(k) \geq C_{\text{weak}}(k) \geq C_{\text{split}}(k, k, k) = 2^{k-2} + 1,$$

where the equality follows from Theorem 6. Moreover, if $C_{\text{weak}}(k) \leq 2^{k-2} + 1$, then the Erdős–Szekeres Conjecture is true.

We do not have $C_{\text{strong}}(k) \leq 2^{k-2} + 1$ as Balko and Valtr [4] proved $C_{\text{strong}}(7) > 33$ with SAT solvers, disproving a conjecture of Peters and Szekeres [19]. Our SAT-solver-based proofs show that this is also the case for weak k -gons.

► **Theorem 7.** *We have $C_{\text{weak}}(7) > 33$.*

Thus, there is no hope for extending the Erdős–Szekeres Conjecture to the abstract setting with weak k -gons. This also suggests that the Erdős–Szekeres Conjecture might be false and that one might look for counterexamples. A natural candidate for a counterexample is a subset of the point sets that were used by Erdős–Szekeres to prove the tightness in Theorem 1 for a -caps and u -cups since such a set used by Erdős–Szekeres [7] contains a subset of 2^{k-2} points with no k points in convex position.

However, we prove that it is impossible to find a counterexample to Erdős–Szekeres Conjecture by restricting ourselves to these sets. In fact, we show this for a larger class of so-called *decomposable sets* (see Section 4 for their definition). These sets include, for example, the point sets with no long caps nor cups used by Erdős–Szekeres [7] as well as the *Horton sets* defined by Valtr [23].

► **Theorem 8.** *Let a, u, k be positive integers satisfying $a, u \leq k$. Then, every decomposable set of more than $\sum_{i=k-a+2}^u \binom{k-2}{i-2}$ points contains an a -cap, a u -cup, or k points in convex position.*

In particular, since every non-empty subset of a decomposable set is decomposable, we obtain, by choosing $a = u = k$, that every subset of more than 2^{k-2} points from a decomposable set contains k points in convex position. Therefore, if we restrict ourselves to decomposable sets, the Erdős–Szekeres Conjecture is true. The proof of Theorem 8 is omitted.

Finally, when exploring the validity of the Erdős–Szekeres Conjecture, we discovered new constructions of sets of 2^{k-2} points in general position with no k points in convex position. As our last main result, we introduce a “blow-up” method that generalizes all previously known constructions of such point sets; see Section 4 for their brief description. Our new method is captured by Lemma 14 in Section 6. We show that all previously known examples follow rather easily from this lemma and that it also gives new examples of such point sets.

We also tried to computationally tackle the Erdős–Szekeres conjecture for large k using a non-linear optimization solver based on a genetic algorithm and this new construction. However, despite finding numerous new examples of 2^{k-2} points without k points in convex position even for $k = 20$, our experiments never produced such points sets with more than 2^{k-2} points; see Section 6 for more details.

3 Open problems

We showed that the variant of the Erdős–Szekeres Conjecture is true for split k -gons, even in an abstract combinatorial setting, while the original Erdős–Szekeres Conjecture does not admit such an abstract strengthening; these results are summarized in Table 1. Each cell corresponds to a variant statement of the Erdős–Szekeres Conjecture.

■ **Table 1** Variants of the Erdős–Szekeres Conjecture and their status.

	Point sets	Ordered hypergraphs
Split k -gons	True (Theorem 6)	True (Theorem 3)
Weak k -gons	Open (Erdős–Szekeres [7])	False (Theorem 7)

It also follows from Theorem 7 that any proof of the Erdős–Szekeres Conjecture has to use some assumptions on the 2-colorings. Natural candidates for such 2-colorings are so-called *signotopes* that appear under many different names in the literature (see [2, 10], for example). For $N \in \mathbb{N}$, a 2-coloring χ of $\mathcal{K}_N^3$ is a *signotope* if for every 4-tuple $a < b < c < d$ of vertices of $\mathcal{K}_N^3$ there is at most one change of color in the sequence $(\chi(a, b, c), \chi(a, b, d), \chi(a, c, d), \chi(b, c, d))$.

Signotopes on $\mathcal{K}_N^3$ are in one-to-one correspondence with pseudoline arrangements determined by N points [10] and thus offer a natural and frequently used combinatorial generalization of point sets. Moreover, if we have $\chi(a, b, c) = \chi(b, c, d)$ for some vertices $a < b < c < d$ in a signotope χ , then $\chi(a, b, c) = \chi(a, b, d) = \chi(a, c, d) = \chi(b, c, d)$. It follows that any weak k -gon in χ is also a strong k -gon in χ .

In our computer experiments, we were not able to find a signotope on at least $1 + \sum_{i=k-a+2}^u \binom{k-2}{i-2}$ vertices that would contain neither a red copy of $\mathcal{P}_a^3$, nor a blue copy of $\mathcal{P}_u^3$, nor a weak k -gon. Thus, it is still possible that the Erdős–Szekeres Conjecture is true even if we do not restrict ourselves to 2-colorings induced by point sets but we consider all signotopes. Such a generalization is equivalent to a conjecture of Goodman and Pollack [12].

Another natural open problem is to obtain better bounds on $C_{weak}(k)$ and $C_{strong}(k)$. We know that both numbers can be larger than $2^{k-2} + 1$, but the strongest upper bound that we are aware of follows from Theorem 5 for $a = u = k$ and gives

$$C_{weak}(k) \leq C_{strong}(k) \leq \binom{2k-4}{k-2} + 1.$$

4 Preliminaries and notation

In this section, we introduce some useful notation and definitions and briefly describe all known constructions of sets of 2^{k-2} that avoid subsets of k points in convex position. Some of these constructions are used later in our proofs.

For a point p in the plane, we use $x(p)$ and $y(p)$ to denote the x - and y -coordinate of p , respectively. Let A and B be two sets of points in the plane in general position. We write $x(A) < x(B)$ (or $y(A) < y(B)$) if A is to the left (or below, respectively) of B , that is, $x(a) < x(b)$ ($y(a) < y(b)$, respectively) for every $a \in A$ and $b \in B$. We say that A is *deep below* B if $y(A) < y(B)$ and all points from B lie above every line determined by two points from A and all points from A lie below every line determined by two points from B .

For integers $a, u \geq 2$, we now describe the sets of $\binom{a+u-4}{a-2}$ points without a -caps and u -cups that were used by Erdős and Szekeres [7] to prove the tight lower bound in Theorem 1. We denote these sets by $P(a, u)$ and we construct them by induction on $a + u$. If $a = 2$ or $u = 2$, then we let $P(a, u)$ be a set of one point. For $a, u \geq 3$, we let $P(a, u)$ be the point set that is obtained by adding a translated copy of $P(a, u - 1)$ to $P(a - 1, u)$ that is to the left of $P(a - 1, u)$ and deep below $P(a - 1, u)$. It is then easy to show that $P(a, u)$ does not contain a -cap nor a u -cup [7, 15].

A set P is *decomposable* if either $|P| = 1$ or if $|P| \geq 2$ and P can be partitioned into two decomposable sets A and B such that A is deep below B . Note that the sets $P(a, u)$ are decomposable and that every non-empty subset of a decomposable set is decomposable.

Let a, u , and k be integers satisfying $(a, u, k) \in T$. We use $S(a, u, k)$ to denote the sum $\sum_{i=k-a+2}^u \binom{k-2}{i-2}$. For $a, u, k \in \mathbb{N}$ with $a, u \leq k$, we also define $S(a, u, k) = 0$ if $a = 1$ or $u = 1$. These cases agree with the formula $\sum_{i=k-a+2}^u \binom{k-2}{i-2}$ as the sum is then empty.

We now describe all known constructions of $S(a, u, k)$ points in general position without a -caps, u -cups, or k points in general position. Let Q be an arbitrary set of $a + u - k - 1$ points $q_{k-a+2}, \dots, q_u$ in the plane in general position sorted by increasing y -coordinate. Let P be the point set obtained by replacing each q_i with a set Q_i of $\binom{k-2}{i-1}$ points without $(k - i + 2)$ -cap and i -cup in a sufficiently small neighborhood of q_i such that Q_i is deep below Q_{i+1} for every $i \in \{k - a + 2, \dots, u - 1\}$. Then, it can be shown that P does not contain a -cap, u -cup, nor k points in convex position [8, 9]. We call the resulting point set *Valtr's construction* as it was described to us by Valtr [24]. To our knowledge, it is not described in the literature. If, in addition, Q is a $(a + u - k - 1)$ -cup and each Q_j is the set $P(k + 1 - j, j + 1)$, then we call the set P the *Erdős-Szekeres construction* and we denote it by $P(a, u, k)$. For $a = u = k$, this is the point set constructed by Erdős and Szekeres [8] to prove $ES(k) \geq 2^{k-2} + 1$.

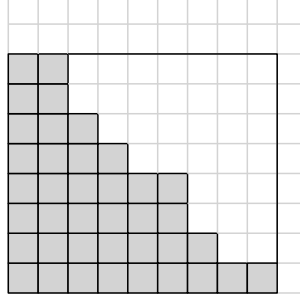
5 Proofs of Theorems 4 and 6

Here, we prove Theorems 4 and 6 by determining the values $ES_{split}(a, u, k)$ and $C_{split}(a, u, k)$ for every $(a, u, k) \in T$. It follows from (3) that in order to prove both these results, it suffices to prove the upper bound $C_{split}(a, u, k) \leq 1 + \sum_{i=k-a+2}^u \binom{k-2}{i-2}$ from Theorem 6 and the lower bound $ES_{split}(a, u, k) \geq 1 + \sum_{i=k-a+2}^u \binom{k-2}{i-2}$ from Theorem 4.

5.1 Upper bound from Theorem 6

Let m and n be positive integers. We use $\preceq$ to denote the partial ordering on $[m] \times [n]$ satisfying $(x_1, y_1) \preceq (x_2, y_2)$ if and only if $x_1 \leq x_2$ and $y_1 \leq y_2$ for all $(x_1, y_1), (x_2, y_2) \in [m] \times [n]$. A subset D of $[m] \times [n]$ is a *down-set* if $y \in D$ implies $x \in D$ for every $x \preceq y$; see Figure 2.

There is a one-to-one correspondence between down-sets in $[m] \times [n]$ and $m \times n$ *line partitions* with entries from $\{0, 1, \dots, n\}$ which are sequences $(a_1, \dots, a_m)$ of numbers satisfying $n \geq a_1 \geq a_2 \geq \dots \geq a_m \geq 0$; see [17]. It is well-known and easy to see that the number of such line partitions (and thus also the number of down-sets in $[m] \times [n]$) is $\binom{m+n}{m}$.



■ **Figure 2** An example of a down-set in $[10] \times [10]$ with the bounding box 9×8 .

Given a non-empty down-set D in $[m] \times [n]$, we say that D has the *bounding box* $r \times s$ if r is the maximum integer such that $a_r > 0$ and $a_1 = s$. We say that the empty down-set has the bounding box 0×0 .

► **Lemma 9.** *For $(a, u, k) \in T$, the number of down-sets in $[a-2] \times [u-2]$ with the bounding box $r \times s$ where $r + s \leq k-1$ is*

$$\sum_{i=k-a+2}^u \binom{k-2}{i-2}.$$

The proof of Lemma 9 is omitted. We now prove the upper bound on $C_{split}(a, u, k)$.

► **Lemma 10.** *For every $(a, u, k) \in T$, we have*

$$C_{split}(a, u, k) \leq 1 + \sum_{i=k-a+2}^u \binom{k-2}{i-2}.$$

Proof. For a positive integer N , let χ be a 2-coloring of $\mathcal{K}_N^3$ with colors red and blue such that there is neither a red copy of $\mathcal{P}_a^3$, a blue copy of $\mathcal{P}_u^3$, nor an abstract split k -gon. We show that $N \leq \sum_{i=k-a+2}^u \binom{k-2}{i-2}$.

For vertices u and v of $\mathcal{K}_N^3$ with $u < v$, we let $D(u, v) = (n_r - 1, n_b - 1)$ where n_r (n_b , respectively) is the maximum number of vertices of a red (blue, respectively) monotone path ending with u and v . By the choice of χ , we have $D(u, v) \in [a-2] \times [u-2]$. Next, we define

$$D(v) = \{x \in [a-2] \times [u-2] : x \preceq D(u, v) \text{ for some } u < v\}.$$

By definition, $D(v)$ is a down-set in $[a-2] \times [u-2]$. Moreover, since there is no abstract split k -gon in χ , $D(v)$ has the bounding box $r \times s$ where $r + s \leq k-1$.

Consider the mapping $v \mapsto D(v)$. We show that this mapping is injective. Suppose for contradiction that $D(u) = D(v)$ for some vertices $u < v$. By definition, we have $D(u, v) \in D(v)$. Since $D(u) = D(v)$, we obtain $D(u, v) \in D(u)$ and thus, by the definition of $D(u)$, there is a vertex t such that $t < u$ and $D(u, v) \preceq D(t, u)$. However, if $\chi(t, u, v)$ is red, the longest red monotone path ending at u, v is longer than the longest red monotone path ending at t, u . An analogous claim holds for blue monotone paths if $\chi(t, u, v)$ is blue. In either case, we cannot have $D(u, v) \preceq D(t, u)$ and we obtain a contradiction.

Thus, N is at most as large as the number of down-sets in $[a-2] \times [u-2]$ with the bounding box $r \times s$ where $r + s \leq k-1$. By Lemma 9, this number equals $\sum_{i=k-a+2}^u \binom{k-2}{i-2}$. ◀

By combining Lemma 10 with (3), we obtain the upper bounds from Theorems 4 and 6.

5.2 Lower bound from Theorem 4

Now, we prove the lower bounds. First, we provide a general construction of 2-colorings that avoid long monochromatic monotone paths and abstract split k -gons, obtaining the lower bound $C_{split}(a, u, k) \geq 1 + \sum_{i=k-a+2}^u \binom{k-2}{i-2}$ from Theorem 6. Then, we show that a special case of this construction can be realized by points in the plane, obtaining a tight lower bound $ES_{split}(a, u, k) \geq 1 + \sum_{i=k-a+2}^u \binom{k-2}{i-2}$ from Theorem 4.

▶ **Lemma 11.** *For every $(a, u, k) \in T$, we have*

$$C_{split}(a, u, k) \geq 1 + \sum_{i=k-a+2}^u \binom{k-2}{i-2}.$$

Proof. We use $\mathcal{E} := \mathcal{D}(a, u)$ to denote the set of all down-sets in $[a-2] \times [u-2]$. In this proof, we let $\mathcal{D} := \mathcal{D}(a, u, k)$ be the set of all down-sets in $[a-2] \times [u-2]$ with the bounding box $r \times s$ where $r + s \leq k-1$ and we set $N = |\mathcal{D}|$. Note that $\mathcal{D} \subseteq \mathcal{E}$. We know, by Lemma 9, that $N = \sum_{i=k-a+2}^u \binom{k-2}{i-2}$. The set $\mathcal{E}$ is partially ordered by inclusion $\subseteq$. We let $\leq_{\mathcal{E}}$ be an arbitrary linear extension on $\mathcal{E}$ that extends $\subseteq$ on $\mathcal{E}$.

For two down-sets $A, B \in \mathcal{E}$ with $A <_{\mathcal{E}} B$, choose an arbitrary element $\delta(A, B)$ of $B \setminus A$. This is possible, as the fact that $\leq_{\mathcal{E}}$ extends $\subseteq$ implies that the set $B \setminus A$ is non-empty if $A <_{\mathcal{E}} B$. For $i \in \{1, 2\}$, we let $\delta(A, B)_i$ be the i th coordinate of $\delta(A, B)$.

We let $\mathcal{K}_{|\mathcal{E}|}^3$ be the ordered complete 3-uniform hypergraph with the vertex set $\mathcal{E}$ ordered by $\leq$. We now define the following 2-coloring $\chi_{a,u} = \chi_{a,u}(\delta, \leq_{\mathcal{E}})$ of $\mathcal{K}_{|\mathcal{E}|}^3$, which is used later in the paper. The color $\chi_{a,u}(A, B, C)$ of an edge $\{A, B, C\}$ of $\mathcal{K}_{|\mathcal{E}|}^3$ with $A < B < C$ is red if $\delta(A, B)_1 < \delta(B, C)_1$ and blue otherwise. Note that $\delta(A, B) \in B$ and $\delta(B, C) \notin B$ and thus $\delta(A, B) \neq \delta(B, C)$. Also, if $\chi_{a,u}(A, B, C)$ is blue, then $\delta(A, B)_2 < \delta(B, C)_2$ as we cannot have $\delta(B, C) \preceq \delta(A, B)$, because otherwise the fact that B is a down-set in $[a-2] \times [u-2]$ together with $\delta(A, B) \in B$ implies $\delta(B, C) \in B$, which is impossible as $\delta(B, C) \in C \setminus B$. Let $\chi = \chi(\delta, \leq_{\mathcal{E}})$ be the coloring $\chi_{a,u}$ restricted to the copy of $\mathcal{K}_N^3$ in $\mathcal{K}_{|\mathcal{E}|}^3$ induced by $\mathcal{D}$.

It remains to show that χ contains neither a red copy of $\mathcal{P}_a^3$, a blue copy of $\mathcal{P}_u^3$, nor an abstract split k -gon as then we have $C_{split}(a, u, k) \geq N + 1$. To do so, we first prove by induction on $\ell \geq 2$ that any monochromatic copy of $\mathcal{P}_{\ell}^3$ in χ with last two vertices $B < C$ satisfies $\delta(B, C)_1 \geq \ell - 1$ if all its hyperedges are red in χ and $\delta(B, C)_2 \geq \ell - 1$ if they are blue. If $\ell = 2$, then the claim is trivially true as $(1, 1) \preceq \delta(B, C)$. Thus, consider $\ell \geq 3$. Let $A < B < C$ be the last three vertices of the monochromatic copy of $\mathcal{P}_{\ell}$ in χ . If all the hyperedges are red, then the induction hypothesis and the definition of χ give $\delta(B, C)_1 > \delta(A, B)_1 \geq \ell - 2$. Similarly, if the hyperedges are blue, we obtain $\delta(B, C)_2 > \delta(A, B)_2 \geq \ell - 2$. Since all down-sets from $\mathcal{D}$ are contained in $[a-2] \times [u-2]$, it follows that there is no red copy of $\mathcal{P}_a^3$ nor a blue copy of $\mathcal{P}_u^3$ in χ . Since every down-set from $\mathcal{D}$ has the bounding box $r \times s$ where $r + s \leq k-1$, we also see that there is no abstract split k -gon in χ . Thus, we have $C_{split}(a, u, k) \geq N + 1$. ◀

We now prove $ES_{split}(a, u, k) \geq N + 1$. This can be shown directly using the Erdős–Szekeres construction $P(a, u, k)$ from Section 4. We prove this claim by showing that a special case of the construction from the proof of Lemma 11 corresponds to $\chi_{P(a,u,k)}$.

► **Lemma 12.** *For every $(a, u, k) \in T$, we have*

$$ES_{split}(a, u, k) \geq 1 + \sum_{i=k-a+2}^u \binom{k-2}{i-2}.$$

The proof of Lemma 12 is omitted. Finally, we remark that using the Erdős–Szekeres construction in the proof of the tight lower bound on $ES_{split}(a, u, k)$ is important. For example, in the case $a = u = k$, some of the sets P of 2^{k-2} points constructed in Section 4 do not have k points in convex position but contain split $(2k - 4)$ -gons.

6 Avoiding k points in convex position

Here, we introduce new constructions of sets of 2^{k-2} points in general position with no k points in convex position, generalizing all previously known constructions of such point sets.

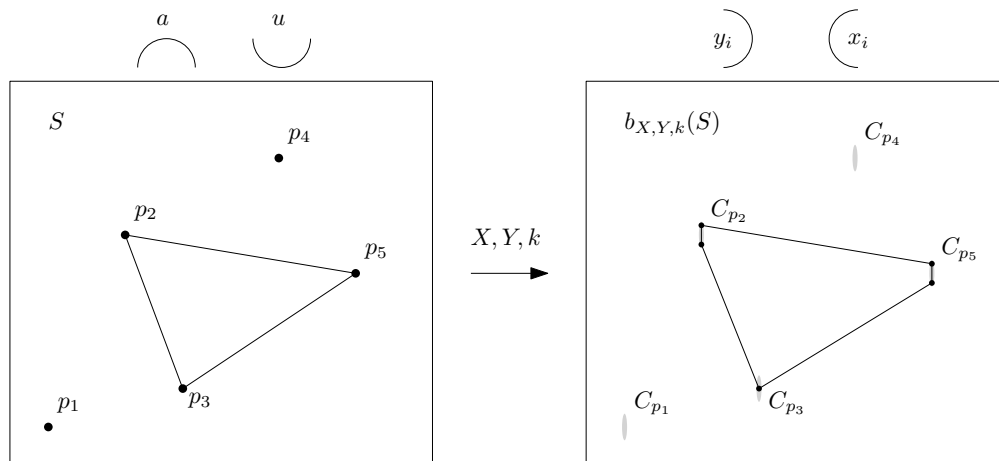
For integers $a, u, k \geq 2$, let $P'(a, u)$ be an arbitrary set of $\binom{a+u-4}{a-2}$ points in general position in the plane with no a -cap and no u -cup. If $(a, u, k) \in T$, let $P'(a, u, k)$ and $P(a, u, k)$ be sets of $\sum_{i=k-a+2}^u \binom{k-2}{i-2}$ points in general position in the plane with no a -cap, no u -cup, and no k points in convex position where $P(a, u, k)$ is the point configuration given by the Erdős–Szekeres construction and $P'(a, u, k)$ is an arbitrary such point configuration. Both sets are described in Section 4.

We let S be a set of points $p_1, \dots, p_N$ sorted by increasing x -coordinates. We use the term *factor* for an integer k that is used as a parameter that corresponds to the forbidden size of a subset in convex position of an enlarged point set obtained from S . For integers i and j with $1 \leq i < j \leq N$, we let $s_{i,j}$ be the maximum number of points from S that are in convex position and have p_i and p_j as their leftmost and rightmost points, respectively.

We now define a blow-up operation that is the core of our construction and will help us obtain large sets with few points in convex position; see Figure 3 for an illustration.

► **Definition 13** ((X, Y) -blow-up of S). *Let $X = (x_1, \dots, x_N)$ and $Y = (y_1, \dots, y_N)$ be sequences of non-negative integers. For a factor $k \in \mathbb{N}_0$, an (X, Y) -blow-up $b_{X,Y,k}(S)$ of S with respect to k is a point set obtained from S by replacing each point p_i of S with $i \in \{2, 3, \dots, N-1\}$ with an affine image C_{p_i} of $P'(y_i+2, x_i+2)$ that is rotated clockwise by $\pi/2$, is almost vertical and lies in a small neighborhood of p_i . The point p_1 is replaced with an affine image C_{p_1} of $P'(k, x_1+2, k)$, where this copy is rotated clockwise by $\pi/2$, is almost vertical and lies in a small neighborhood of p_1 . Similarly, the point p_N is analogously replaced with an affine image C_{p_N} of $P'(y_N+2, k, k)$. We choose the neighborhoods small enough so that they are disjoint for different sets C_{p_i} and for all triples i_1, i_2, i_3 with $1 \leq i_1 < i_2 < i_3 \leq N$, the triple $(p_{i_1}, p_{i_2}, p_{i_3})$ has the same orientation as each triple $(q_{i_1}, q_{i_2}, q_{i_3})$ where $q_{i_j} \in C_{p_{i_j}}$. Also, C_{p_i} is almost vertical, meaning that for every $p_j \in S$ with $j \neq i$ any line determined by two points of C_{p_i} has all points of C_{p_j} on the same side as the vertical line containing p_i . We call the sets C_{p_i} clusters.*

For each p_i , every a -cap in the pre-image of C_{p_i} corresponds to a *left convex chain* of size a in C_{p_i} and every u -cup in the pre-image of C_{p_i} corresponds to a *right convex chain* of size u in C_{p_i} . We note that the constructions could be stated without the rotation by $\pi/2$, but this version will be more natural in later constructions. In the following result, we state how (X, Y) -blows-ups of arbitrary point sets affect the size of the largest subset in convex position in the resulting point sets and also describe how large these point sets are.



■ **Figure 3** An illustration of an (X, Y) -blow-up of a point set S and its subset in convex position.

► **Lemma 14** (General blow-up construction). *For $k, N \in \mathbb{N}$, and a set S of N points in the plane in general position, let $X = (x_1, \dots, x_N)$ and $Y = (y_1, \dots, y_N)$ be sequences of non-negative integers satisfying*

- (a) $x_i + y_i \leq k - 1$ for every $i \in [N]$ and
- (b) $x_i + y_j \leq k - 1 - s_{i,j}$ for all $i, j \in [N]$ with $i < j$.

Then, the (X, Y) -blow-up $b_{X,Y,k}(S)$ of S does not contain k points in convex position and

$$|b_{X,Y,k}(S)| = \sum_{l=0}^{x_1} \binom{k-2}{l} + \sum_{i=2}^{N-1} \binom{x_i + y_i}{x_i} + \sum_{l=0}^{y_N} \binom{k-2}{l}.$$

The proof of Lemma 14 is omitted. Observe that it follows from the second condition that the factor k is larger than the size of the largest convex subset of S as the numbers x_i and y_i must be non-negative.

6.1 All known constructions

We now show that for each set S and factor $k = |S| + 1$ there are sequences X and Y such that $|b_{X,Y,k}(S)| = 2^{k-2}$ with no k points in convex position. It will follow that all known constructions of sets of 2^{k-2} points in general position with no k points in convex position are a special case of an (X, Y) -blow-up. We recall that all such known constructions fall within Valtr's construction. Now, let S be an arbitrary set of $N \leq k - 1$ points in general position. We set $x_i = i - 1$, and $y_i = k - i - 1$ for every $i \in [N]$. Then, clearly, $x_i + y_i = k - 2 \leq k - 1$ and $x_i, y_i \geq 0$ for every $i \in [N]$ as $|S| = N \leq k - 1$. We have $x_i + y_j \leq k - 1 - s_{i,j}$ for all $i, j \in [N]$ with $i < j$, as $s_{i,j} \leq j - i + 1$ and $(i - 1) + (k - j - 1) = k - 1 - (j - i + 1) \leq k - 1 - s_{i,j}$. Thus, all assumptions in Lemma 14 are met, and $b_{X,Y,k}(S)$ does not contain k points in convex position. If $N = k - 1$, then we also get

$$\begin{aligned} |b_{X,Y,k}(S)| &= \sum_{l=0}^0 \binom{k-2}{l} + \sum_{i=2}^{N-1} \binom{x_i + y_i}{x_i} + \sum_{l=0}^0 \binom{k-2}{l} \\ &= 1 + \sum_{i=2}^{k-2} \binom{k-2}{i-1} + 1 = 2^{k-2} \end{aligned}$$

and $b_{X,Y,k}(S)$ corresponds to Valtr's construction rotated by $\pi/2$.

6.2 New construction

The previous choices of X and Y are not the only ones that work, we describe another choice of an (X, Y) -blow-up to obtain a set of 2^{k-2} points with no k -tuple in convex position for any $k \geq 3$. This time, the sequence X will not be increasing, and we allow $k < |S|$.

We now proceed with the construction. For each point p_i , each subset of S in convex position with p_i as its rightmost point (leftmost point) is called a *left-gon at p_i* (*right-gon at p_i* , respectively). For a *threshold* $L \in \{0, 1, \dots, N\}$, we call points p_i with $i \in [L]$ *left* and points p_j with $j \in [N] \setminus [L]$ *right*. For each left (right, respectively) point p_i , we define s_i as the maximum size of left-gon at p_i (right-gon at p_i , respectively). Note that if $L \in [N-1]$ there are exactly two points p_i of S with $s_i = 1$, namely, the leftmost and the rightmost point of S . For every positive integer j , let $v_{j,L}(S)$ be the number of points p_i of S with $s_i = j$.

If $x_i = x$ and $y_i = k - 1 - x - s_i$ for every $i \in [L]$ and $x_i = k - 1 - x - s_i$ and $y_i = x$ for each $i \in [N] \setminus [L]$, then we call the set $b_{X,Y,k}(S)$ an *x -blow-up of S with respect to threshold L and factor k* and denote it by $b_{x,k,L}(S)$. We now show that the size of the x -blow-up of S can be expressed in terms of the numbers $v_{j,L}(S)$.

► **Corollary 15.** *For a set S of N points in the plane in general position, let x be a non-negative integer and choose $m = k - 2x$ for some threshold $L \in [N-1]$ and factor $k \geq 1$. If S does not contain m points in convex position, then the x -blow-up $b_{x,k,L}(S)$ of S does not contain k points in convex position and*

$$|b_{x,k,L}(S)| = 2 \sum_{l=0}^x \binom{k-2}{l} + \sum_{j=2}^{k-2x-1} v_{j,L}(S) \cdot \binom{k-j-1}{x}.$$

The proof of Corollary 15 is omitted. It follows from Corollary 15 that in order to produce large x -blow-ups without k points in convex position, point sets S with suitable values $v_{j,L}(S)$ for some threshold L are useful. We now provide a simple formula for the values $v_{j,N}(P)$ in the Erdős–Szekeres construction $P = P(k, k, k)$ where all points are labeled as left.

► **Lemma 16.** *For every integer $k \geq 2$, if we set the threshold $L = 2^{k-2}$, then, for $P = P(k, k, k)$, we have*

$$v_{j,L}(P) = \begin{cases} 1, & \text{for } j = 1, \\ 2^{j-2}, & \text{for } j \in \{2, \dots, k-1\}, \\ 0, & \text{otherwise.} \end{cases}$$

The proof of Lemma 16 is omitted. Analogously, we get the same values $v_{j,0}(P)$ if all points of $P(k, k, k)$ are labeled as right, that is, if we set $L = 0$.

We now construct a point set M with $|M| = 2^{m-2}$ and without m points in convex position that has larger x -blow-ups than $P = P(m, m, m)$.

► **Definition 17 (Point set M).** *We construct the following set M that contains $2m-2$ points $\ell_1, \dots, \ell_{m-1}, r_1, \dots, r_{m-1}$ in the plane in general position. We let $\ell_1 = (-m, 0)$, $r_1 = (m, \varepsilon)$, $\ell_2 = (-m+1, 1)$, and $r_2 = (m-1, 1+\varepsilon)$ for some small $\varepsilon > 0$. We now specify the placement of the remaining points $\ell_3, \dots, \ell_{m-1}$ and $r_3, \dots, r_{m-1}$ inductively; see Figure 4. Assume that we have placed points $\ell_1, \dots, \ell_i$ and $r_1, \dots, r_i$ for some $i \geq 2$. We now let ℓ_{i+1} be a point above the lines $\overline{\ell_{i-1}\ell_i}$ and $\overline{r_{i-1}r_i}$ to the right of ℓ_{i-1} and to the left of the line $x = 0$ and we let r_{i+1} be a point above the lines $\overline{\ell_{i-1}\ell_i}$ and $\overline{r_{i-1}r_i}$ to the left of r_i and to the right of the line $x = 0$ so that $y(r_{i+1}) > y(\ell_{i+1})$.*

We let P' be a point set obtained by replacing each point ℓ_i with $i \in [m-1]$ with a copy of the set $P(m-i+1, i)$ that lies in a sufficiently small neighborhood of ℓ_i . We denote this copy as C_{ℓ_i} . Note that $C_{\ell_1} = \emptyset$. We then let P'' be a point set obtained by replacing each point r_i with $i \in [m-1]$ with a copy of the set $P(m-i, i+1)$ that lies in a sufficiently small neighborhood of r_i . We use C_{r_i} to denote this copy and note that $C_{r_{m-1}} = \emptyset$. Note that the point sets P' and P'' then form affine copies of the point set $P(m-1, m-1, m-1)$. Let $M = P' \cup P''$ and note that $|M| = 2^{m-2}$.

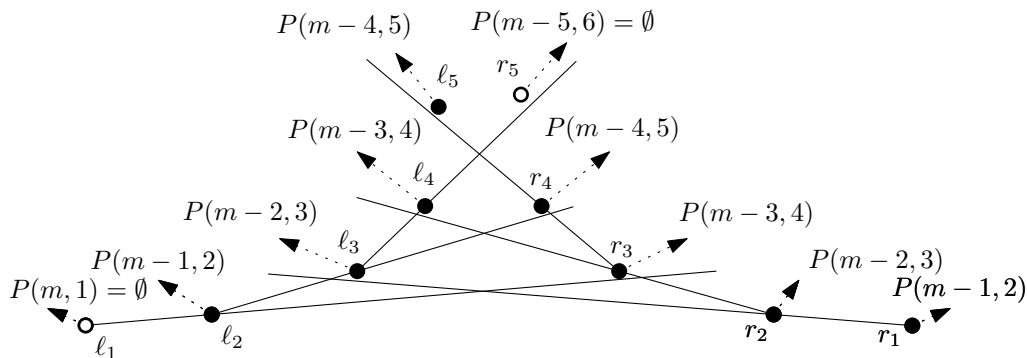


Figure 4 Example of the construction of the point set M for $m = 6$.

Now, if we set the threshold $L = 2^{m-3}$, then the left points of M are the points from P' , and the right points of M are the points of P'' . We also obtain the following formulas.

► **Lemma 18.** For every integer $m \geq 3$, the set M does not contain m points in convex position and, for the threshold $L = 2^{m-3}$, satisfies

$$v_{j,L}(M) = \begin{cases} 2, & \text{for } j = 1, \\ 2^{j-1}, & \text{for } j \in \{2, \dots, m-2\}, \\ 0, & \text{otherwise.} \end{cases} \quad (4)$$

The proof of Lemma 18 is omitted. We now prove that suitable x -blow-ups of M produce new sets of 2^{k-2} points in the plane in general position that do not contain k points in convex position, obtaining a new construction of such point sets.

► **Theorem 19.** For each positive integer x and $k \geq 2x + 3$, let $m = k - 2x$. The x -blow-up $b_{x,k,L}(M)$ of the point set M with respect to the threshold $L = 2^{m-3}$ does not contain k points in convex position and $|b_{x,k,L}(M)| = 2^{k-2}$.

Proof. By applying the x -blow-up from Corollary 15 with parameters x , k , and $L = 2^{m-3}$ to M , we obtain a set $b_{x,k,L}(M)$ of

$$|b_{x,k,L}(M)| = \sum_{l=0}^x \binom{k-2}{l} + \sum_{j=2}^{k-2x-2} 2^{j-1} \cdot \binom{k-j-1}{x} + \sum_{l=k-x-2}^{k-2} \binom{k-2}{l}$$

points in the plane in general position that contains no k points in convex position.

To show that $|b_{x,k,L}(M)|$ equals 2^{k-2} it suffices to prove that

$$\sum_{l=x+1}^{k-x-3} \binom{k-2}{l} = \sum_{j=2}^{k-2x-2} 2^{j-1} \cdot \binom{k-j-1}{x}$$

if $k \geq 2x + 3$, as then $|b_{x,k,L}(S)| = 2^{k-2}$ follows by the binomial theorem. This equality is true because both sides count the number of subsets of $[k-2]$ of size at least $x+1$ and at most $k-x-3$. This is trivial for the left-hand side. To prove the right-hand side, let $\mathcal{F}$ be the set of subsets of $[k-2]$ of size at least $x+1$ and at most $k-x-3$. For $F \in \mathcal{F}$, we let j be the minimum element from $[k-2]$ such that either F or its complement contains exactly $x+1$ elements in $\{j-1, \dots, k-2\}$. Note that such j is uniquely determined for each $F \in \mathcal{F}$ and that $j \in \{2, \dots, k-2x-2\}$. On the other hand, we can uniquely determine set F from $\mathcal{F}$ by selecting $j-1$, deciding whether F or its complement has exactly $x+1$ elements in $\{j-1, \dots, k-2\}$ and then select the remaining x of them from $\{j, \dots, k-2\}$ to F (or to its complement, respectively) and, finally, select an arbitrary subset of $[j-2]$ to F (or its complement, respectively). ◀

6.3 Computational experiments

Since there are other possible choices for X and Y , one might wonder whether some of them yield constructions of more than 2^{k-2} points without k points in convex position. We implemented a simple program in *Matlab* based on a non-linear optimization solver to find as large (X, Y) -blow-up of a given point set as possible. Given a set of N points in the plane in general position and a factor k , this solver tries to find a solution $(x_1, \dots, x_N, y_1, \dots, y_N)$ that satisfies the linear constraints from Lemma 14 and that has as large value of the function

$$\sum_{l=0}^{x_1} \binom{k-2}{l} + \sum_{i=2}^{N-1} \binom{x_i + y_i}{x_i} + \sum_{l=0}^{y_N} \binom{k-2}{l}$$

as possible. Since the objective function is not linear, we used a solver based on a genetic algorithm that does not guarantee finding a true optimum solution. We tried various point sets on the input, including subsets of $P(a, u)$ and random point sets. Using this program, we obtained many examples of 2^{k-2} points for various choices of S and k up to 20. Even though there is some freedom in choosing X and Y , the size of the (X, Y) -blow-up never exceeded 2^{k-2} . So the Erdős–Szekeres conjecture remains open.

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